

Multicollinearity in Logistic Regression: Theory, Consequences, and Application

*A Project report submitted in partial fulfilment of the
requirements for the Degree of M.Sc. (Statistics)
with specialization in Industrial Statistics*

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CERTIFICATE

This is to certify that **Mr. Mahajan Ghansham Dnyaneshwar, Miss. Maheshwari Ishwarlal Patil, Mr. Pardeshi Aditya Kishorsing** students of M.Sc.(Statistics) with specialization in Industrial Statistics, at Kavayitri Bahinabai Chaudhari, North Maharashtra University, Jalgaon have successfully completed their project work entitled "Multicollinearity in Logistic Regression: Theory, Consequences, and Application" as a part of M.Sc. (Statistics) program under my guidance and supervision during the academic year 2025-2026.

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Abstract

Multicollinearity can seriously affect parameter estimation in logistic regression models by causing instability and inflated Mean Squared Error (MSE). This study investigates the impact of multicollinearity on logistic regression and examines the performance of several biased and shrinkage estimators as alternatives to the classical Maximum Likelihood Estimator(MLE).

A Monte Carlo simulation study is conducted under different levels of correlation, sample sizes, and numbers of explanatory variables to compare the estimators using the MSE criterion. In addition, a real prostate cancer dataset is analyzed to illustrate the practical consequences of multicollinearity.

The results from both the simulation and real data analysis indicate that shrinkage-based estimators provide more stable and efficient estimates than the MLE when multicollinearity is present.

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Chapter 1

Introduction and Literature Review

1.1 Introduction

Logistic regression is one of the most widely used statistical techniques for modeling binary response variables in medical, biological, and social science research. It establishes a relationship between a binary outcome variable and a set of explanatory variables through a nonlinear link function. Due to its interpretability and flexibility, logistic regression has become a standard tool in applied statistical analysis.

However, the presence of multicollinearity among explanatory variables poses a serious challenge in logistic regression modeling. Multicollinearity occurs when two or more predictors are highly correlated, leading to instability in parameter estimation, inflated variances, and unreliable statistical inference. In such situations, the classical Maximum Likelihood Estimator (MLE) may perform poorly, producing large standard errors and high Mean Squared Error (MSE).

To address the adverse effects of multicollinearity, several biased and shrinkage estimation techniques have been proposed in the literature. These estimators introduce a small amount of bias into the estimation process in order to achieve a substantial reduction in variance, thereby improving overall estimation accuracy. While such methods have been extensively studied in linear regression models, their application and performance in logistic regression settings require further investigation.

The primary objective of this study is to examine the performance of various biased and shrinkage logistic regression estimators in the presence of multicollinearity. A comprehensive comparison is carried out using theoretical results, Monte Carlo simulation experiments, and a real-life prostate cancer dataset. The performance of the estimators is evaluated using the Mean Squared Error criterion under different levels of correlation, sample

sizes, and model dimensions. The findings of this study aim to provide practical guidance on the selection of appropriate logistic regression estimators when multicollinearity is present.

1.2 Literature Review

Multicollinearity is a well-known problem in regression analysis as it leads to unstable parameter estimates and inflated variances. In linear regression, [Hoerl and Kennard \[1981\]](#) proposed the ridge estimator to reduce the effect of multicollinearity by introducing a small bias to achieve variance reduction. Later, [Kejian \[1993\]](#) introduced the Liu estimator, which provides additional flexibility in controlling the shrinkage effect.

These ideas were subsequently extended to logistic regression, where multicollinearity adversely affects the Maximum Likelihood Estimator (MLE), resulting in large standard errors and high Mean Squared Error (MSE). Several biased and shrinkage estimators, including ridge-type, Liu-type, and modified ridge estimators, were developed to improve estimation accuracy in logistic regression models.

More recent studies have proposed Stein-type shrinkage estimators, such as the Logistic Stein Ridge Estimator (LSRE) and Logistic Stein Kibria Estimator (LSKE), which have been shown to perform well under severe multicollinearity. However, comprehensive comparative studies covering a wide range of estimators and different levels of multicollinearity are limited. The present study aims to address this gap by comparing several biased logistic regression estimators using simulation experiments and areal data application.

Motivation of the Study

The motivation of the present study can be summarized as follows:

- Multicollinearity among explanatory variables has a significant adverse effect on parameter estimation in logistic regression models, leading to unstable estimates and inflated variances.
- In the presence of strong correlation among predictors, the classical Maximum Likelihood Estimator (MLE) becomes unreliable and may result in poor estimation accuracy.
- Biased and shrinkage estimation methods are commonly employed to address this issue by introducing a small amount of bias in exchange for a substantial reduction in variance.

- The primary motivation of this study is to examine how such biasing methods can effectively reduce the impact of multicollinearity on parameter estimates in logistic regression models.
- Accordingly, this work focuses on comparing various biased logistic regression estimators to assess their ability to provide more stable and reliable parameter estimates under multicollinearity.

1.2.1 Diagnostic Approaches

Pearson correlation coefficient

Usually we use Pearson's correlation coefficient to measure the strength of the association between two variables. The general rule of thumb is that if correlation coefficient between two variables is greater than 0.8 or 0.9, the multicollinearity is a serious problem. Formula of sample correlation coefficient is described as follows.

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

r is sample correlation coefficient, n is sample size, x_i , y_i are the individual sample points indexed with i is mean of $\bar{x}$ variable and is mean of $\bar{y}$ variable

Tolerance

Furthermore, multicollinearity can mainly be detected with the help of tolerance and its reciprocal, called variance inflation factor (VIF). The tolerance is the percentage of the variance in a given predictor that cannot be explained by the other predictors.

By definition tolerance of any specific explanatory variable is

$$Tolerance = 1 - R^2$$

where R^2 is the coefficient of determination for the regression of that explanatory variable on all remaining independent variables. Tolerance close to 1 indicates that there is little multicollinearity, whereas a value close to zero suggests that multicollinearity may be a threat. There is no formal cut-off value to use with tolerance for determining presence of multicollinearity [Midi et al. \[2010\]](#). [Myers Mayers \[1990\]](#) suggests a tolerance value below 0.1

indicates serious collinearity problem and Menard [Menard \[2001\]](#) suggests that a tolerance value less than 0.2 indicates a potential collinearity problem. As a rule of thumb, a tolerance of 0.1 or less is a cause for concern

VIF

The VIF is defined as the reciprocal of tolerance as

$$VIF = \frac{1}{TOLERANCE}$$

VIF shows that how much the variance of the coefficient estimate is being inflated by multicollinearity. Like tolerance there is no formal cutoff value to use with VIF for determining the presence of multicollinearity. Values of VIF exceeding 10 are often regarded as indicating multicollinearity.

Eigen values, condition index and variance proportions

Moreover, eigen values for the scaled, uncentered cross-product matrix, condition indices and the variance proportions for each explanatory variable is used to identify multicollinearity. If any eigen value is larger than others, then of the regression parameters can be greatly affected by small changes in the explanatory variables or outcome. If the eigen values are fairly similar then the fitted model is likely to be unchanged by small changes in the measured variables [Rana et al. \[2010\]](#)

The condition indices are computed as the square root of the ratio of the largest eigen value to the eigen value of interest. It is defined as

$$K = \sqrt{\frac{\lambda_{max}}{\lambda_k}}$$

where λ_{max} and λ_k are the maximum and the k th eigen values respectively. When there is no collinearity at all, the eigen values, condition indices will equal unity. As collinearity increases, eigen values will be both greater and smaller than unity. If one or more of the eigenvalues are small (close to zero) and the corresponding condition number is large, then we have an indication of multicollinearity. There is no hard and fast rule about how much larger a condition index needs to be indicated collinearity problems. An informal rule of thumb is that if the condition index is 15, multicollinearity is a concern; if it is greater than 30, multicollinearity is a very serious concern [Midi et al. \[2010\]](#)

Table 1.1: Description of Variables in the Prostate Cancer Dataset

Variable	Description
y	Seminal vesicle invasion (1 = presence, 0 = absence)
x_1	Cancer volume
x_2	Patient weight
x_3	Age of the patient
x_4	Benign prostate hyperplasia
x_5	Capsular penetration
x_6	Gleason score
x_7	Prostate-specific antigen (PSA) level

Table 1.2: Eigenvalues of the Matrix $X'WX$ for the Prostate Cancer Dataset

Eigenvalue	Value
λ_1	69109.31254
λ_2	58.42260
λ_3	35.33693
λ_4	13.64340
λ_5	10.92500
λ_6	5.07500
λ_7	2.50087

Data Description and Problem Statement

To assess multicollinearity among the explanatory variables, the eigenvalues of the matrix $X'WX$ were computed. A large disparity between the largest and smallest eigenvalues is observed, indicating near-linear dependence among the regressors. The condition index, defined as $\sqrt{\lambda_{\max}/\lambda_{\min}}$, is calculated as 166.2351, which is far greater than the commonly used threshold value of 100. This confirms the presence of severe multicollinearity in the prostate cancer dataset. As a result, classical maximum likelihood estimation is expected to be unstable, making the dataset suitable for evaluating biased and shrinkage logistic regression estimators.

Chapter 2

Estimation Methods

In this Chapter, we will focus on the logistic regression model and explore various types of estimators, their bias, covariance, and MSE.

2.1 Logistic Regression Using Maximum Likelihood Estimation

We consider the following logistic regression model:

$$y_i = \pi_i + \varepsilon_i, \quad i = 1, 2, \dots, n, \quad (2.1)$$

where ε_i are disturbances with

$$E(\varepsilon_i) = 0 \quad \text{and} \quad \text{Cov}(\varepsilon_i) = \pi_i(1 - \pi_i).$$

and π_i is the expectation of y_i , given by

$$\pi_i = \frac{\exp(x_i' \underline{\beta})}{1 + \exp(x_i' \underline{\beta})}, \quad (2.2)$$

where x_i is the i -th row of X in an $n \times (p+1)$ data matrix with p explanatory (independent) variables, and $\underline{\beta}$ is a $(p+1) \times 1$ vector of coefficients.

The maximum likelihood estimator (MLE) of $\underline{\beta}$ maximizes the log-likelihood function:

$$l(\underline{\beta}) = \sum_{i=1}^n \left[y_i \log(\pi_i) + (1 - y_i) \log(1 - \pi_i) \right]. \quad (2.3)$$

Setting the first derivative of (2.3) equal to zero gives the score equation:

$$\frac{\partial l}{\partial \underline{\beta}} = \sum_{i=1}^n (y_i - \pi_i) x_i = 0. \quad (2.4)$$

Since this equation is nonlinear in β , it can be solved using the Newton-Raphson method. Alternatively, the following Iteratively Reweighted Least Squares (IRLS) algorithm can be used:

$$\hat{\beta}_{\text{MLE}} = (X' \hat{W} X)^{-1} X' \hat{W} z, \quad (2.5)$$

where $\hat{W} = \text{diag}[\hat{\pi}_i(1 - \hat{\pi}_i)]$ is the weight (variance) matrix and z is the working dependent variable with i -th element

$$z_i = \log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) + \frac{y_i - \hat{\pi}_i}{\hat{\pi}_i(1 - \hat{\pi}_i)}. \quad (2.6)$$

Let Q be the orthogonal matrix whose columns are the eigenvectors of $X' \hat{W} X$. Then the asymptotic covariance matrix of the MLE is given by the inverse of the Hessian:

$$\text{Cov}(\hat{\beta}_{\text{MLE}}) = E\left[-\frac{\partial^2 l}{\partial \beta \partial \beta'}\right]^{-1} = (X' \hat{W} X)^{-1}. \quad (2.7)$$

The matrix mean squared error (MMSE) and scalar mean squared error (MSE) of the MLE are

$$\text{MMSE}(\hat{\beta}_{\text{MLE}}) = Q \Lambda^{-1} Q', \quad (2.8)$$

$$\text{MSE}(\hat{\beta}_{\text{MLE}}) = E[(\hat{\beta}_{\text{MLE}} - \beta)'(\hat{\beta}_{\text{MLE}} - \beta)] = \text{tr}(Q \Lambda^{-1} Q') = \sum_{j=1}^{p+1} \frac{1}{\lambda_j}, \quad (2.9)$$

where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_{p+1})$ and λ_j are the eigenvalues of $X' \hat{W} X$.

It can be easily seen that the asymptotic variance becomes inflated when the independent variables are highly correlated, because some of the eigenvalues will be small when $X' \hat{W} X$ is ill-conditioned. If the degree of collinearity is severe, one or more of the estimates will be imprecise and may not reflect the true effects of the independent variables. In practice, multicollinearity inflates the estimated variances and causes precision problems when identifying the effects of the independent variables.

2.2 Logistic Ridge Regression

In the research paper by [Schaefer et al. \[1984\]](#), a Ridge estimator was presented for use in logistic regression when the independent variables are severely collinear.

The Ridge logistic estimator is given by

$$\hat{\beta}_{\text{LRE}} = (X' \hat{W} X + kI)^{-1} X' \hat{W} X \hat{\beta}_{\text{MLE}} \quad (2.10)$$

where k is called the *biasing parameter* ($k > 0$) and I is the identity matrix. The bias, covariance matrix, and MMSE of the estimator are given by

$$\text{Bias}(\hat{\beta}_{\text{LRE}}) = -kQ\Lambda_k^{-1}\alpha, \quad (2.11)$$

$$\text{Cov}(\hat{\beta}_{\text{LRE}}) = Q\Lambda_k^{-1}\Lambda\Lambda_k^{-1}Q', \quad (2.12)$$

$$\text{MMSE}(\hat{\beta}_{\text{LRE}}) = Q\Lambda_k^{-1}\Lambda\Lambda_k^{-1}Q' + k^2Q\Lambda_k^{-1}\alpha\alpha'\Lambda_k^{-1}Q', \quad (2.13)$$

where

$$\Lambda_k = \text{diag}(\lambda_1 + k, \lambda_2 + k, \dots, \lambda_{p+1} + k), \quad (2.14)$$

and $\alpha = Q'\beta$. Then the asymptotic MSE of the logistic ridge regression estimator is

$$\text{MSE}(\hat{\beta}_{\text{LRE}}) = \text{tr}(\text{MMSE}(\hat{\beta}_{\text{LRE}})) = \sum_{j=1}^{p+1} \frac{\lambda_j}{(\lambda_j + k)^2} + \sum_{j=1}^{p+1} \frac{k^2 \alpha_j^2}{(\lambda_j + k)^2}. \quad (2.15)$$

Now, the asymptotic variance is not inflated because λ_j is replaced by $\lambda_j + k$ in the denominator. As the asymptotic variance decreases, the squared bias becomes large at the same time as k increases. Therefore, the objective in logistic ridge regression is to choose a value of k such that the reduction in the variance term is greater than the increase in the squared bias.

The Ridge Logistic Estimator will have a smaller MSE than $\hat{\beta}_{\text{MLE}}$ for some k when the data are sufficiently multicollinear and the sample size n is sufficiently large. From the empirical study in the paper, the findings indicate that the Ridge Estimator has a smaller MSE when the sample size is approximately 250 and the degree of multicollinearity is moderate to high.

Rules to Choose the Ridge Parameter k

In ridge logistic regression, the biasing parameter k controls the amount of shrinkage applied to the estimated coefficients. Selecting an appropriate value for k is crucial for balancing bias and variance and achieving a lower mean squared error (MSE) than standard maximum likelihood estimation, especially under multicollinearity.

- **Rule I:**

$$k = \frac{1}{\max \left(\hat{\lambda}'_j \hat{\beta}_{\text{MLE}} \right)^2}, \quad (2.16)$$

where $\hat{\lambda}_j$ are the eigenvalues of $X'WX$.

- **Rule II:**

$$k = \frac{p+1}{\hat{\beta}'_{\text{MLE}} \hat{\beta}_{\text{MLE}}}, \quad (2.17)$$

where p is the number of independent variables and $\hat{\beta}_{\text{MLE}}$ are their maximum likelihood estimates.

- **Rule III:**

$$k = 0.1, \quad (2.18)$$

which is a simple, more conservative (i.e., smaller) choice for k .

Empirical studies indicate that Rule II generally outperforms Rule I and Rule III in reducing MSE, no matter the sample size or how strongly the variables are correlated.

2.3 Liu-Type Logistic Estimator

In the presence of multicollinearity, the maximum likelihood estimator (MLE) in the logistic regression model can exhibit high variance and unstable coefficient estimates. To address this issue, biased estimation methods such as the ridge and Liu estimators have been proposed. Building on this idea, Kejian [Kejian \[1993\]](#), Urgan and Tez [Urgan and Tez \[2008\]](#), and Matin et al. [Matin et al. \[2005\]](#) introduced the Liu-type logistic estimator (LLE), which has the advantage that its estimates are a linear function of the shrinkage parameter d , simplifying its selection. Due to this advantage over the ridge estimator, the Liu-type estimator has been widely used in the literature to mitigate multicollinearity in logistic regression models.

The Liu-type estimator is an alternative to the ridge estimator and accounts for multicollinearity in both the linear regression model (LRM) and the logistic regression model. The main advantage of $\hat{\beta}_{\text{LLE}}$ over $\hat{\beta}_{\text{LRE}}$ is that it is linear in d , making the choice of d simpler.

Definition

The Liu-type logistic estimator is defined as:

$$\hat{\beta}_{\text{LLE}} = (X' \hat{W} X + I)^{-1} (X' \hat{W} X - dI) \hat{\beta}_{\text{MLE}} ; 0 < d < 1, \quad (2.19)$$

where I is the identity matrix and $\hat{\beta}_{\text{MLE}}$ is the maximum likelihood estimator. When $d = 1$, the LLE reduces to the MLE:

$$\hat{\beta}_{\text{LLE}}|_{d=1} = \hat{\beta}_{\text{MLE}}$$

Bias, Covariance, and MMSE

Let Q be the orthogonal matrix whose columns are the eigenvectors of $X' \hat{W} X$, and let $\alpha = Q' \beta$. Define:

$$\Lambda_I = \text{diag}(\lambda_1 + 1, \lambda_2 + 1, \dots, \lambda_{p+1} + 1), \quad (2.20)$$

$$\Lambda_d = \text{diag}(\lambda_1 + d, \lambda_2 + d, \dots, \lambda_{p+1} + d). \quad (2.21)$$

The bias vector, covariance matrix, and matrix mean squared error (MMSE) of the Liu-type estimator are given by

$$\text{Bias}(\hat{\beta}_{\text{LLE}}) = -(1 - d) Q \Lambda_I^{-1} \alpha, \quad (2.22)$$

$$\text{Cov}(\hat{\beta}_{\text{LLE}}) = Q \Lambda_I^{-1} \Lambda_d \Lambda_I^{-1} Q^T, \quad (2.23)$$

$$\text{MMSE}(\hat{\beta}_{\text{LLE}}) = Q \Lambda_I^{-1} \Lambda_d \Lambda_I^{-1} Q^T + (1 - d)^2 Q \Lambda_I^{-1} \alpha \alpha^T \Lambda_I^{-1} Q^T, \quad (2.24)$$

where Q is the orthogonal matrix of eigenvectors of $X^T \hat{W} X$, Λ is the diagonal matrix of corresponding eigenvalues, and $\alpha = Q^T \beta$. The scalar mean squared error (MSE) of the Liu-type estimator is

$$\text{MSE}(\hat{\beta}_{\text{LLE}}) = \sum_{j=1}^{p+1} \frac{(\lambda_j + d)^2}{\lambda_j (\lambda_j + 1)^2} + \sum_{j=1}^{p+1} \frac{(1 - d)^2 \alpha_j^2}{(\lambda_j + 1)^2}, \quad (2.25)$$

where λ_j denotes the j -th eigenvalue of $X^T \hat{W} X$ and α_j is the j -th element of α .

Performance in Presence of Multicollinearity

When multicollinearity exists, $X' \hat{W} X$ is nearly singular, resulting in inflated variance for the MLE:

$$\text{Var}(\hat{\beta}_{\text{MLE}}) = (X' \hat{W} X)^{-1}$$

The LLE addresses this issue as follows:

$$\text{Var}(\hat{\beta}_{\text{LLE}}) = (X' \hat{W} X + kI)^{-1} (X' \hat{W} X - dI) \text{Var}(\hat{\beta}_{\text{MLE}}) (X' \hat{W} X - dI) (X' \hat{W} X + kI)^{-1}$$

- Variance is reduced due to the ridge parameter k .
- Bias introduced by d is moderate and controlled.
- Mean squared error (MSE) is often smaller than that of the MLE.
- Estimates are more stable and reliable under multicollinearity.

2.4 The Almost Unbiased Ridge Estimator

Kadiyala [1984] assumed that $\hat{\underline{\beta}}$ is a biased estimator of $\underline{\beta}$, with the bias vector defined as

$$\text{Bias}(\hat{\underline{\beta}}) = E(\hat{\underline{\beta}}) - \underline{\beta} = A\underline{\beta},$$

which implies that $E(\hat{\underline{\beta}}) - A\underline{\beta} = \underline{\beta}$. In this case, the estimator

$$\tilde{\underline{\beta}} = \hat{\underline{\beta}} - A\hat{\underline{\beta}} = (I - A)\hat{\underline{\beta}}$$

is called the *almost unbiased estimator* based on the biased estimator $\hat{\underline{\beta}}$.

An *Almost Unbiased Ridge Estimator* (AURE) for the logistic regression model, based on the ridge estimator of Schaefer Schaefer et al. [1984], was proposed by Chang Chang [2015] and can be written as:

$$\hat{\underline{\beta}}_{\text{AURE}} = (I - k^2(X'WX + kI)^{-2})\hat{\underline{\beta}}_{\text{MLE}} \quad (2.26)$$

The bias, covariance matrix, MMSE, and scalar MSE of the estimator are, respectively, given by:

$$\text{Bias}(\hat{\underline{\beta}}_{\text{AURE}}) = -k^2\Lambda_k^{-2}\alpha, \quad (2.27)$$

$$\text{Cov}(\hat{\underline{\beta}}_{\text{AURE}}) = Q(I - k^2\Lambda_k^{-2})\Lambda^{-1}(I - k^2\Lambda_k^{-2})Q' \quad (2.28)$$

$$\text{MMSE}(\hat{\underline{\beta}}_{\text{AURE}}) = Q(I - k^2\Lambda_k^{-2})\Lambda^{-1}(I - k^2\Lambda_k^{-2})Q' + k^4Q\Lambda_k^{-2}\alpha\alpha'\Lambda_k^{-2}Q' \quad (2.29)$$

$$\text{MSE}(\hat{\underline{\beta}}_{\text{AURE}}) = \sum_{j=1}^{p+1} \frac{(\lambda_j + 2k^2)\lambda_j}{(\lambda_j + k)^4} + \sum_{j=1}^{p+1} \frac{k^4\alpha_j^2}{(\lambda_j + k)^4} \quad (2.30)$$

Here, Q is the orthogonal matrix of eigenvectors of $X'WX$, $\Lambda_k = \text{diag}(\lambda_1 + k, \lambda_2 + k, \dots, \lambda_{p+1} + k)$, and $\alpha = Q'\underline{\beta}$.

2.5 The Almost Unbiased Liu Estimator (AULE)

The Almost Unbiased Liu Estimator (AULE) is an improvement over the classical Liu estimator in logistic regression models, particularly when multicollinearity is present. While the standard Liu estimator reduces variance

by shrinking the maximum likelihood estimator (MLE) using the shrinkage parameter d , it introduces bias. The AULE reduces this bias while preserving the variance reduction advantages.

Kadiyala [1984] proposed the concept of an almost unbiased estimator: if $\hat{\beta}$ is a biased estimator of β with

$$\text{Bias}(\hat{\beta}) = E(\hat{\beta}) - \beta = A\beta$$

then the almost unbiased estimator is

$$\tilde{\beta} = \hat{\beta} - A\hat{\beta} = (I - A)\hat{\beta}$$

Based on this, the Almost Unbiased Liu Estimator for logistic regression Månsson et al. [2012], Chang [2015] is defined as

$$\hat{\beta}_{\text{AULE}} = [I - (1 - d)^2(X'WX + I)^{-2}] \hat{\beta}_{\text{MLE}} ; 0 < d < 1 \quad (2.31)$$

where I is the identity matrix and $\hat{\beta}_{\text{MLE}}$ is the maximum likelihood estimator. When $d = 1$, the AULE reduces to the MLE:

$$\hat{\beta}_{\text{AULE}}|_{d=1} = \hat{\beta}_{\text{MLE}}$$

Bias, Covariance, and MMSE

Let Q be the orthogonal matrix of eigenvectors of $X'WX$, and $\alpha = Q'\beta$. Define

$$\Lambda_I = \text{diag}(\lambda_1 + 1, \lambda_2 + 1, \dots, \lambda_{p+1} + 1) \quad (2.32)$$

$$\Lambda_d = \text{diag}(\lambda_1 + d, \lambda_2 + d, \dots, \lambda_{p+1} + d) \quad (2.33)$$

Then, the bias vector, covariance matrix, and mean squared error (MMSE) of the AULE are

$$\text{Bias}(\hat{\beta}_{\text{AULE}}) = -(1 - d)^2 Q \Lambda_I^{-2} \alpha \quad (2.34)$$

$$\text{Cov}(\hat{\beta}_{\text{AULE}}) = Q (I - (1 - d)^2 \Lambda_I^{-2}) \Lambda^{-1} (I - (1 - d)^2 \Lambda_I^{-2}) Q' \quad (2.35)$$

$$\text{MMSE}(\hat{\beta}_{\text{AULE}}) = \text{Cov}(\hat{\beta}_{\text{AULE}}) + (1 - d)^4 Q \Lambda_I^{-2} \alpha \alpha' \Lambda_I^{-2} Q' \quad (2.36)$$

The scalar mean squared error (MSE) is

$$\text{MSE}(\hat{\beta}_{\text{AULE}}) = \sum_{j=1}^{p+1} \frac{(\lambda_j + d)^2 (2 + \lambda_j - d)^2}{\lambda_j (\lambda_j + 1)^4} + \sum_{j=1}^{p+1} \frac{(1 - d)^4 \alpha_j^2}{(\lambda_j + 1)^4} \quad (2.37)$$

2.6 Logistic K-L Estimator (LKLE)

The Kibria Logistic Estimator (KLE) was proposed by [Lukman et al. \[2023\]](#) as an alternative biased estimator for the logistic regression model under multicollinearity. The K-L estimator belongs to the class of ridge-type and Liu-type estimators with a single biasing parameter. It is a form of the Liu-type estimator that minimizes the residual sum of squares with respect to the L_2 norm while incorporating prior information. The K-L estimator often outperforms the Ridge Regression Estimator (RRE) and Liu Estimator (LE) under theoretical conditions.

Logistic K-L Estimator (LKLE)

In logistic regression, the Logistic K-L estimator (LKLE) is defined as

$$\hat{\beta}_{\text{LKLE}} = (X' \hat{W} X + k I_p)^{-1} (X' \hat{W} X - k I_p) \hat{\beta}_{\text{MLE}} \quad (2.38)$$

where $\hat{W}$ is the weight matrix from the logistic regression model evaluated using $\hat{\beta}_{\text{MLE}}$

Bias, Variance, and Mean Squared Error

The bias of the LKLE is

$$\text{Bias}(\hat{\beta}_{\text{LKLE}}) = -2kQ\Lambda_k\alpha \quad (2.39)$$

where

$$\Lambda_k = (\Lambda + k I_p)^{-1}, \quad \alpha = Q' \beta \quad (2.40)$$

and Q is the orthogonal matrix of eigenvectors of $X' \hat{W} X$, Λ is the diagonal matrix of eigenvalues.

The variance of the LKLE is

$$\text{Cov}(\hat{\beta}_{\text{LKLE}}) = Q(\Lambda - k I_p) \Lambda_k \Lambda^{-1} \Lambda_k (\Lambda - k I_p) Q' \quad (2.41)$$

The matrix mean squared error (MMSE) and scalar mean squared error (MSE) are given by

$$\text{MMSE}(\hat{\beta}_{\text{LKLE}}) = Q(\Lambda - k I_p) \Lambda_k \Lambda^{-1} \Lambda_k (\Lambda - k I_p) Q' + 4k^2 Q \Lambda_k \alpha \alpha' \Lambda_k Q' \quad (2.42)$$

$$\text{MSE}(\hat{\beta}_{\text{LKLE}}) = \sum_{j=1}^{p+1} \frac{(\lambda_j - k)^2}{\lambda_j (\lambda_j + k)^2} + 4k^2 \sum_{j=1}^{p+1} \frac{\alpha_j^2}{(\lambda_j + k)^2} \quad (2.43)$$

2.7 The Adjusted Logistic Liu Estimator (ALLE)

The Adjusted Logistic Liu Estimator (ALLE) is a modification of the Liu estimator, introduced by [Amin et al. \[2023\]](#), to improve estimation in logistic regression models affected by multicollinearity. To address multicollinearity, commonly used biased estimators, such as the Ridge estimator (RE) and Liu estimator (LE), are preferred. However, a major drawback of the traditional LE is that its Liu parameter (LP) can sometimes attain negative values. To overcome this issue, the Adjusted Liu Estimator (ALLE) has been proposed for the binary logistic regression model. It is defined as:

$$\hat{\beta}_{\text{ALLE}} = (X' \hat{W} X + I)^{-1} (X' \hat{W} X - dI) \hat{\beta}_{\text{MLE}} \quad (2.44)$$

where d is the biasing parameter.

The bias vector, covariance matrix, MMSE, and scalar MSE of $\hat{\beta}_{\text{ALLE}}$ are given by:

$$\text{Bias}(\hat{\beta}_{\text{ALLE}}) = -(1 + d)Q^{-1}\alpha \quad (2.45)$$

$$\text{Cov}(\hat{\beta}_{\text{ALLE}}) = Q^{-1}(-dI)^{-1}(-dI)^{-1}IQ \quad (2.46)$$

$$\text{MMSE}(\hat{\beta}_{\text{ALLE}}) = Q^{-1}(-dI)^{-1}(-dI)^{-1}IQ + (1 + d)^2 Q^{-1}\alpha\alpha^T IQ \quad (2.47)$$

$$\text{MSE}(\hat{\beta}_{\text{ALLE}}) = \sum_{j=1}^{p+1} \frac{(\lambda_j - d)^2}{\lambda_j(\lambda_j + 1)^2} + \sum_{j=1}^{p+1} \frac{(1 + d)^2 \alpha_j^2}{(\lambda_j + 1)^2} \quad (2.48)$$

2.8 The Dorugade Logistic Estimator (LDE)

The Dorugade Logistic Estimator (LDE) is a biased shrinkage estimator proposed as an alternative to the Maximum Likelihood Estimator (MLE) in logistic regression models affected by multicollinearity. It introduces two biasing parameters, k and d , to balance bias and variance.

The Dorugade Logistic Estimator is defined as:

$$\hat{\beta}_{\text{LDE}} = (X' \hat{W} X + kdI)^{-1} (X' \hat{W} X) \hat{\beta}_{\text{MLE}} ; k > 0, 0 < d < 1 \quad (2.49)$$

Let $X' \hat{W} X = Q\Lambda Q'$ be the eigen-decomposition, with Λ diagonal and Q orthogonal, and $\alpha = Q^T \beta$.

Bias, Variance, and MMSE

$$\text{Bias}(\hat{\beta}_{\text{LDE}}) = -kdQ(\Lambda + kdI)^{-1}\alpha \quad (2.50)$$

$$\text{Cov}(\hat{\beta}_{\text{LDE}}) = Q(\Lambda + kdI)^{-1}\Lambda(\Lambda + kdI)^{-1}Q^\top \quad (2.51)$$

$$\begin{aligned} \text{MMSE}(\hat{\beta}_{\text{LDE}}) &= Q(\Lambda + kdI)^{-1}\Lambda(\Lambda + kdI)^{-1}Q^\top \\ &\quad + k^2d^2Q(\Lambda + kdI)^{-1}\alpha\alpha^\top(\Lambda + kdI)^{-1}Q^\top \end{aligned} \quad (2.52)$$

Scalar Mean Squared Error (MSE)

$$\text{MSE}(\hat{\beta}_{\text{LDE}}) = \sum_{j=1}^{p+1} \frac{\lambda_j}{(\lambda_j + kd)^2} + \sum_{j=1}^{p+1} \frac{k^2d^2\alpha_j^2}{(\lambda_j + kd)^2} \quad (2.53)$$

Choice of Parameters Biasing parameter k :

- Hoerl and Kennard (1975): $k = \hat{\sigma}^2 / (\hat{\beta}_{\text{MLE}}^\top \hat{\beta}_{\text{MLE}})$
- Kibria (2003): $k = \text{Median}_i (\hat{\sigma}^2 \lambda_i / \hat{\alpha}_i^2)$
- Cross-validation can also be used.

Biasing parameter d :

$$d^* = \frac{\sum_{i=1}^p \lambda_i(\lambda_i + k)\hat{\alpha}_i^2 / (\lambda_i + kd)^2}{\sum_{i=1}^p \lambda_i^2 / (\lambda_i + kd)^2} \quad (2.54)$$

Special Cases

- $k = 0, d = 0 \Rightarrow$ LDE reduces to MLE
- $d = 1 \Rightarrow$ LDE reduces to Logistic Ridge Estimator

2.9 Modified Ridge-Type Estimator for Logistic Regression (MRT)

The Modified Ridge-Type (MRT) estimator is a two-parameter ridge-family shrinkage estimator adapted for logistic regression by Lukman et al. [2020]. It generalizes the logistic ridge estimator by introducing an additional parameter d to improve bias-variance tradeoff.

Eigen-space Representation

Let $X^\top W X = Q\Lambda Q^\top$ be the spectral decomposition with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{p+1})$ and Q orthogonal. Define $\alpha = Q^\top \beta$. Then

$$\hat{\alpha}_j^{\text{MRT}} = \frac{\lambda_j}{\lambda_j + k(1+d)} \hat{\alpha}_j^{\text{MLE}}, \quad j = 1, \dots, p+1 \quad (2.55)$$

$$\hat{\underline{\beta}}_{MRT} = Q \operatorname{diag} \left(\frac{\lambda_j}{\lambda_j + k(1+d)} \right) Q' \hat{\underline{\beta}}_{MLE} \quad (2.56)$$

Bias:

$$\operatorname{Bias}(\hat{\underline{\beta}}_{MRT}) = -Q \operatorname{diag} \left(\frac{k(1+d)}{\lambda_j + k(1+d)} \right) \alpha \quad (2.57)$$

Covariance:

$$\operatorname{Cov}(\hat{\underline{\beta}}_{MRT}) = Q(\Lambda + k(1+d)I)^{-1} \Lambda (\Lambda + k(1+d)I)^{-1} Q' \quad (2.58)$$

Mean Squared Error (MSE)

$$\operatorname{MSE}(\hat{\underline{\beta}}_{MRT}) = \sum_{j=1}^{p+1} \frac{\lambda_j}{(\lambda_j + k(1+d))^2} + \sum_{j=1}^{p+1} \frac{k^2(1+d)^2 \alpha_j^2}{(\lambda_j + k(1+d))^2} \quad (2.59)$$

MMSE of MRT Estimator

Let $\hat{\underline{\beta}}_{MRT}$ be the Modified Ridge-Type estimator and $X' \hat{W} X = Q \Lambda Q'$ its eigen-decomposition with $\alpha = Q' \underline{\beta}$. Then, the MMSE is given by:

$$\operatorname{MMSE}(\hat{\underline{\beta}}_{MRT}) = Q \operatorname{diag} \left(\frac{\lambda_j}{(\lambda_j + k(1+d))^2} \right) Q' + Q \operatorname{diag} \left(\frac{k^2(1+d)^2}{(\lambda_j + k(1+d))^2} \alpha_j^2 \right) Q'$$

where:

- λ_j are the eigenvalues of $X' \hat{W} X$
- α_j are the rotated coordinates, $\alpha = Q' \underline{\beta}$
- $k > 0$ is the ridge parameter, and $d \geq 0$ is the modification parameter.

The first term represents the covariance contribution, while the second term accounts for the squared bias.

Special Cases

- $k = 0, d = 0 \Rightarrow$ MRT reduces to the MLE.
- $d = 0 \Rightarrow$ MRT reduces to standard logistic ridge regression.

Advantages

- Provides flexibility with two tuning parameters for improved bias-variance tradeoff.
- Generalizes the logistic ridge estimator.

- Achieves lower MSE than MLE under multicollinearity.
- Allows more stable coefficient estimation without overshrinking important predictors.

Limitations of the Modified Ridge-Type (MRT) Estimator

- Biased Estimator.
- Dependence on Tuning Parameters.
- Computational Complexity
- Not Always Superior.
- Interpretability.
- Risk of Overshrinkage.

2.10 Logistic Stein Ridge Estimator (LSRE)

Hoque and Kibria [Hoque and Kibria \[2024\]](#) proposed two new Stein-type estimators. One of them is the Stein estimator for the logistic ridge regression estimator, defined as:

$$\hat{\beta}_{LSRE} = c_r \hat{\beta}_{LRE} = c_r A \hat{\beta}_{MLE} \quad ; 0 < c_r < 1 \quad (2.60)$$

where

$$A = (X' \hat{W} X + kI)^{-1} X' \hat{W} X \quad (2.61)$$

The shrinkage factor c_r is selected using a procedure analogous to the linear Stein estimator:

$$c_r = \frac{\hat{\beta}_{MLE}^T A^T A \hat{\beta}_{MLE}}{\hat{\beta}_{MLE}^T A \hat{\beta}_{MLE} + \text{tr}(\text{MSE}(A(X^T \hat{W} X)^{-1} A^T))} \quad (2.62)$$

Bias , covariance matrix, MMSE, and scalar MSE:

$$\text{Bias}(\hat{\beta}_{LSRE}) = (c_r A - I) \alpha, \quad (2.63)$$

$$\text{Cov}(\hat{\beta}_{LSRE}) = c_r^2 Q A \Lambda^{-1} A^T Q^T, \quad (2.64)$$

$$\text{MMSE}(\hat{\beta}_{LSRE}) = c_r^2 Q A \Lambda^{-1} A^T Q^T + Q(c_r A - I) \alpha \alpha^T (c_r A - I)^T Q^T, \quad (2.65)$$

$$\text{MSE}(\hat{\beta}_{LSRE}) = \sum_{j=1}^{p+1} \frac{c_r^2 \lambda_j}{(\lambda_j + k)^2} + \sum_{j=1}^{p+1} \frac{(c_r \lambda_j - \lambda_j - k)^2 \alpha_j^2}{(\lambda_j + k)^2} \quad (2.66)$$

Where:

- Q is the orthogonal matrix of eigenvectors of $X^\top \hat{W} X$,
- $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{p+1})$ is the diagonal matrix of eigenvalues,
- $\alpha = Q^\top \underline{\beta}$
- $k > 0$ is the ridge parameter.

2.11 Stein's Estimator for the K-L Estimator (LSKE)

Hoque and Kibria [2024] proposed two new Stein estimators. The second of them is the Stein estimator for the K-L estimator, defined as:

$$\hat{\underline{\beta}}_{LSKE} = c_k \hat{\underline{\beta}}_{KLRE} = c_k K \hat{\underline{\beta}}_{MLE} \quad ; 0 < c_k < 1, \quad (2.67)$$

where

$$K = (X^\top \hat{W} X + kI)^{-1} (X^\top \hat{W} X - kI) \quad (2.68)$$

The shrinkage factor c_k is chosen as:

$$c_k = \frac{\hat{\underline{\beta}}_{MLE}^\top K \hat{\underline{\beta}}_{MLE}}{\hat{\underline{\beta}}_{MLE}^\top K^\top K \hat{\underline{\beta}}_{MLE} + \text{tr}(\text{MSE}(K(X^\top \hat{W} X)^{-1} K^\top))} \quad (2.69)$$

Bias, Covariance, MMSE and MSE:

$$\text{Bias}(\hat{\underline{\beta}}_{LSKE}) = (c_k K - I) \alpha \quad (2.70)$$

$$\text{Cov}(\hat{\underline{\beta}}_{LSKE}) = c_k^2 Q K \Lambda^{-1} K^\top Q^\top \quad (2.71)$$

$$\text{MMSE}(\hat{\underline{\beta}}_{LSKE}) = c_k^2 Q K \Lambda^{-1} K^\top Q^\top + Q(c_k K - I) \alpha \alpha^\top (c_k K - I)^\top Q^\top \quad (2.72)$$

$$\text{MSE}(\hat{\underline{\beta}}_{LSKE}) = \sum_{j=1}^{p+1} \frac{c_k^2 (\lambda_j - k)^2}{(\lambda_j + k)^2} + \sum_{j=1}^{p+1} \frac{(c_k (\lambda_j - k) - \lambda_j - k)^2 \alpha_j^2}{(\lambda_j + k)^2} \quad (2.73)$$

Where:

- Q is the orthogonal matrix of eigenvectors of $X^\top \hat{W} X$.
- $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{p+1})$ is the diagonal matrix of eigenvalues.
- $\alpha = Q^\top \underline{\beta}$
- $k > 0$ is the K-L biasing parameter.

Selection of Biasing Parameters k and d

The selection of the biasing parameters k and d plays a crucial role in controlling the bias–variance trade-off and aims to minimize the Mean Squared Error (MSE) or prediction error. Several approaches for selecting these parameters have been proposed in the literature. The commonly used methods are summarized below.

- **Hoerl-type rule for k :** Ridge-type heuristics suggest selecting k as

$$k \approx \frac{\hat{\sigma}^2}{\hat{\boldsymbol{\beta}}_{MLE}^\top \hat{\boldsymbol{\beta}}_{MLE}},$$

where $\hat{\sigma}^2$ is replaced by a deviance-based variance estimate in the context of logistic regression. This method is mainly used as a rule-of-thumb and is included here for theoretical reference.

- **Kibria and median-based rules:** These methods make use of the eigenvalues of the matrix $X^\top W X$ and the rotated coefficients $\boldsymbol{\alpha} = Q^\top \boldsymbol{\beta}$ to obtain median-based estimates of the biasing parameters. Although simple to compute, these rules may not always lead to optimal results.
- **Analytic minimization of MSE:** In some cases, the biasing parameters can be obtained by analytically minimizing the MSE. This involves setting the partial derivatives of the MSE with respect to k or d equal to zero and solving the resulting nonlinear equations numerically. Such solutions are estimator-specific and computationally demanding.
- **Cross-validation (method used in this study):** In this study, the biasing parameters k and d are selected using a cross-validation procedure implemented in R. The Hoerl-type, Kibria-based, and analytic minimization rules are presented for theoretical completeness only. The actual selection of the biasing parameters is carried out using cross-validation in order to obtain data-driven and stable estimates. The implementation of this procedure is provided in the Appendix, where the corresponding R function is described in detail.

Bias–Variance Trade-off for the Parameters k and d

The biasing parameters k and d play an important role in controlling the bias–variance trade-off in biased logistic regression estimators. In the presence of multicollinearity, the classical Maximum Likelihood Estimator (MLE)

has zero bias but large variance, which often results in a high Mean Squared Error (MSE).

Introducing a positive value of k or an appropriate value of d shrinks the estimated regression coefficients toward zero. This shrinkage reduces the variance of the estimators but introduces a small amount of bias. As the value of k increases, the variance decreases while the bias increases. A similar behavior is observed for the parameter d , where suitable values help balance bias and variance.

The objective in selecting k and d is to achieve an optimal compromise between bias and variance such that the overall MSE is minimized. In this study, this balance is achieved through a cross-validation approach, which leads to improved estimation accuracy in the presence of multicollinearity.

Chapter 3

Simulation Based Evaluation of Logistic Estimators

3.1 Monte Carlo Simulation

To compare the performance of the estimators, a simulation study has been conducted in this section. We provided an overview of the simulation technique, Simulation results and analyze the simulated data and evaluate the performance of the estimators under different scenarios.

We conducted a Monte Carlo simulation study to examine the performance of different logistic regression estimators. A substantial number of simulations were carried out to compare the performance of these estimators. This simulation procedure follows the approach outlined by McDonald and [McDonald and Galarneau \[1975\]](#) and [Gibbons \[1981\]](#).

To evaluate the performance of the considered estimators under multicollinearity, a Monte Carlo simulation study have conducted following the procedure below:

1. **Data Generation:** The explanatory variables $\underline{X} = (X_1, X_2, \dots, X_p)$ were generated from a multivariate normal distribution

$$\underline{X} \sim N_p(0, \Sigma), \quad \text{with } \Sigma_{ij} = \rho^{|i-j|},$$

where ρ represents the correlation between predictors. Different values of ρ were considered to reflect varying degrees of multicollinearity. The true regression coefficients $\underline{\beta}$ were set according to a pre-specified vector.

2. **Response Variable Generation:** The binary response Y_i was generated from the logistic model

$$P(Y_i = 1 | x_i) = \pi(x_i) = \frac{\exp(x_i^\top \underline{\beta})}{1 + \exp(x_i^\top \underline{\beta})}, \quad i = 1, 2, \dots, n.$$

Conditional on $\pi(x_i)$, $Y_i \sim \text{Bernoulli}(\pi(x_i))$.

3. **Estimator Computation:** For each simulated dataset, coefficient estimates were obtained using all considered estimators, namely MLE, Ridge, Liu, AURE, AULE, ALLE, LKLE, LDE, LMRT, LSRE, and LSKE.
4. **Performance Evaluation:** The Mean Squared Error (MSE) of each estimator was computed as

$$\text{MSE}(\hat{\underline{\beta}}) = \frac{1}{R} \sum_{i=1}^R (\hat{\underline{\beta}}_{ij} - \underline{\beta}_i') (\hat{\underline{\beta}}_{ij} - \underline{\beta}_i')$$

where R denotes the number of replications, $\hat{\beta}_{ij}$ represents the estimate of the i th parameter in the j th replication, and $\underline{\beta}$ denotes the vector of true parameter values. The true parameter vector is selected as the eigenvector of $X^\top W X$ corresponding to the largest eigenvalue and is normalized such that $\underline{\beta}^\top \underline{\beta} = 1$.

5. **Simulation Settings:** The simulation was conducted under varying conditions: sample sizes $n = 30, 50, 100, 200$, number of replications $R = 2000$, and correlation levels $\rho = 0.3, 0.5, 0.7, 0.9$.

This procedure ensures a comprehensive assessment of estimator performance under different degrees of multicollinearity and sample sizes.

Table 3.1: MSE values for selected estimators ($p = 3$)

Estimator	n = 30					n = 50				
	0.7	0.8	0.9	0.95	0.99	0.7	0.8	0.9	0.95	0.99
MLE	1.776	2.530	4.847	9.517	50.682	1.122	1.494	2.795	5.483	28.402
LRE	0.660	0.657	0.597	0.562	0.514	0.608	0.590	0.559	0.516	0.456
LLE	1.088	1.251	1.486	1.844	5.312	0.860	0.969	1.223	1.502	3.362
AURE	0.856	0.865	0.757	0.654	0.486	0.766	0.784	0.767	0.676	0.465
AULE	1.493	1.886	2.612	3.585	11.624	1.050	1.300	1.950	2.762	7.211
LSE	1.201	1.538	2.584	4.575	22.794	0.913	1.087	1.678	2.855	12.876
LKLE	0.868	1.259	3.043	7.315	47.989	0.564	0.638	1.329	3.470	25.723
LDE	1.205	1.478	2.108	3.092	6.884	0.887	1.065	1.533	2.235	5.367
LMRT	0.641	0.639	0.587	0.559	0.520	0.590	0.570	0.541	0.505	0.458
LSRE	0.889	1.013	1.277	1.646	2.602	0.722	0.766	0.931	1.232	2.359
LSKE	1.358	1.841	2.966	4.954	23.170	0.649	1.115	2.058	3.293	13.318

Table 3.2: MSE values for selected estimators ($p = 3$)

Estimator	n = 100					n = 200				
	0.7	0.8	0.9	0.95	0.99	0.7	0.8	0.9	0.95	0.99
MLE	0.666	0.864	1.503	2.725	13.315	0.456	0.548	0.859	1.445	6.522
LRE	0.509	0.528	0.528	0.485	0.392	0.419	0.445	0.487	0.476	0.381
LLE	0.606	0.720	0.961	1.188	2.046	0.444	0.515	0.706	0.918	1.483
AURE	0.588	0.666	0.753	0.716	0.472	0.442	0.507	0.647	0.712	0.548
AULE	0.657	0.834	1.310	1.922	4.312	0.454	0.545	0.828	1.261	2.912
LSE	0.687	0.803	1.129	1.679	6.289	0.546	0.620	0.813	1.104	3.299
LKLE	0.450	0.429	0.520	1.177	10.820	0.400	0.389	0.358	0.443	4.323
LDE	0.596	0.710	1.022	1.435	3.395	0.437	0.500	0.684	0.962	2.384
LMRT	0.496	0.506	0.500	0.461	0.388	0.413	0.431	0.459	0.444	0.367
LSRE	0.584	0.623	0.720	0.849	1.635	0.494	0.524	0.590	0.651	1.094
LSKE	0.430	0.406	1.157	2.106	6.807	0.431	0.401	0.353	1.125	3.853

Table 3.3: MSE values for selected estimators ($p = 5$)

Estimator	n = 30					n = 50				
	0.7	0.8	0.9	0.95	0.99	0.7	0.8	0.9	0.95	0.99
MLE	3.728	5.651	11.515	24.017	128.921	2.143	3.118	6.260	12.602	67.332
LRE	0.714	0.677	0.634	0.589	0.565	0.664	0.648	0.582	0.537	0.477
LLE	1.615	1.813	2.224	2.856	9.801	1.312	1.509	1.836	2.205	5.096
AURE	0.927	0.863	0.746	0.617	0.499	0.895	0.885	0.765	0.641	0.440
AULE	2.602	3.233	4.509	6.308	22.137	1.839	2.355	3.396	4.644	11.975
LSE	1.903	2.663	4.933	9.924	51.383	1.290	1.667	2.898	5.296	26.611
LKLE	2.170	3.717	9.060	21.275	125.832	0.981	1.556	4.091	9.968	64.216
LDE	2.428	3.089	4.679	6.867	15.828	1.608	2.112	3.251	4.850	11.654
LMRT	0.704	0.670	0.631	0.591	0.573	0.651	0.636	0.574	0.535	0.483
LSRE	1.185	1.357	1.871	2.460	3.718	0.881	0.992	1.240	1.695	3.028
LSKE	2.224	3.007	5.270	10.297	51.754	1.586	2.047	3.339	5.734	27.068

Table 3.4: MSE values for selected estimators ($p = 5$)

Estimator	n = 100					n = 200				
	0.7	0.8	0.9	0.95	0.99	0.7	0.8	0.9	0.95	0.99
MLE	1.112	1.606	2.990	5.995	31.089	0.672	0.892	1.567	2.972	14.934
LRE	0.588	0.590	0.550	0.500	0.408	0.502	0.521	0.535	0.491	0.383
LLE	0.899	1.114	1.446	1.803	3.051	0.621	0.760	1.068	1.412	2.243
AURE	0.762	0.827	0.814	0.719	0.442	0.591	0.676	0.787	0.775	0.515
AULE	1.069	1.464	2.273	3.356	7.052	0.666	0.870	1.418	2.235	4.938
LSE	0.874	1.093	1.627	2.804	12.519	0.672	0.786	1.086	1.646	6.236
LKLE	0.511	0.607	1.329	3.690	27.993	0.428	0.407	0.522	1.261	12.004
LDE	0.969	1.279	1.980	3.024	7.557	0.629	0.795	1.233	1.898	5.196
LMRT	0.576	0.574	0.535	0.489	0.408	0.493	0.508	0.516	0.473	0.377
LSRE	0.661	0.725	0.830	1.078	2.251	0.556	0.584	0.658	0.747	1.507
LSKE	0.586	1.147	2.079	3.325	13.066	0.403	0.409	1.112	2.129	6.825

Table 3.5: Estimated MSE values of the selected estimators for $p = 10$

Estimator	n = 30					n = 50				
	0.7	0.8	0.9	0.95	0.99	0.7	0.8	0.9	0.95	0.99
MLE	13.296	21.249	47.932	102.534	595.663	6.047	9.641	20.514	43.396	246.788
LRE	0.761	0.728	0.700	0.679	0.656	0.711	0.684	0.625	0.597	0.565
LLE	2.844	2.932	3.234	3.645	14.139	2.457	2.726	2.971	3.087	3.984
AURE	0.884	0.798	0.698	0.619	0.539	0.912	0.843	0.686	0.584	0.461
AULE	5.530	6.215	7.768	9.489	34.516	4.095	5.044	6.439	7.619	11.467
LSE	5.307	8.178	18.538	38.973	242.419	2.591	3.901	7.670	15.822	91.766
LKLE	11.019	18.754	45.066	99.471	592.407	4.040	7.247	17.715	40.279	243.385
LDE	7.537	10.226	17.007	26.366	67.816	4.430	6.177	10.281	16.224	48.239
LMRT	0.759	0.728	0.701	0.681	0.660	0.707	0.682	0.625	0.599	0.569
LSRE	2.074	2.555	3.923	5.392	8.115	1.347	1.657	2.253	3.335	5.989
LSKE	5.543	8.425	18.804	39.261	242.730	2.911	4.232	8.034	16.185	92.176

Table 3.6: Estimated MSE values of the selected estimators for $p = 10$

Estimator	n = 100					n = 200				
	0.7	0.8	0.9	0.95	0.99	0.7	0.8	0.9	0.95	0.99
MLE	2.634	4.003	8.666	18.173	99.612	1.332	2.001	4.001	8.292	45.473
LRE	0.657	0.625	0.566	0.522	0.459	0.596	0.589	0.549	0.488	0.390
LLE	1.730	2.075	2.641	2.957	2.986	1.109	1.440	2.030	2.585	2.815
AURE	0.916	0.878	0.759	0.621	0.408	0.808	0.861	0.845	0.723	0.422
AULE	2.347	3.140	4.792	6.343	8.392	1.294	1.847	3.067	4.619	7.390
LSE	1.391	1.858	3.556	6.939	36.432	0.927	1.178	1.883	3.419	16.946
LKLE	1.217	2.162	6.098	15.117	96.042	0.548	0.772	2.014	5.616	41.901
LDE	2.263	3.143	5.700	9.804	28.239	1.239	1.747	3.097	5.402	18.067
LMRT	0.652	0.620	0.563	0.521	0.462	0.591	0.582	0.543	0.484	0.391
LSRE	0.849	0.939	1.265	1.852	4.076	0.648	0.692	0.809	1.066	2.513
LSKE	1.788	2.299	4.039	7.429	36.955	0.805	1.454	2.399	3.984	17.560

Simulation results discussion

From all the tables, it is observed that the Mean Squared Error (MSE) increases as the multicollinearity level (ρ) increases, and decreases as the sample size (n) increases. The Maximum Likelihood Estimator (MLE) shows very large MSE values, especially when ρ is high and the number of predictors (p) increases, indicating that MLE performs poorly under multicollinearity. In comparison, shrinkage estimators perform much better. Among all estimators, LRE and LMRT consistently produce the lowest MSE values across different values of ρ , p, and n. Estimators such as LDE and LSRE also show good performance but are generally slightly inferior to LRE and LMRT. Overall, the table results clearly indicate that LRE and LMRT are the best-performing and most reliable estimators for logistic regression when multicollinearity is present.

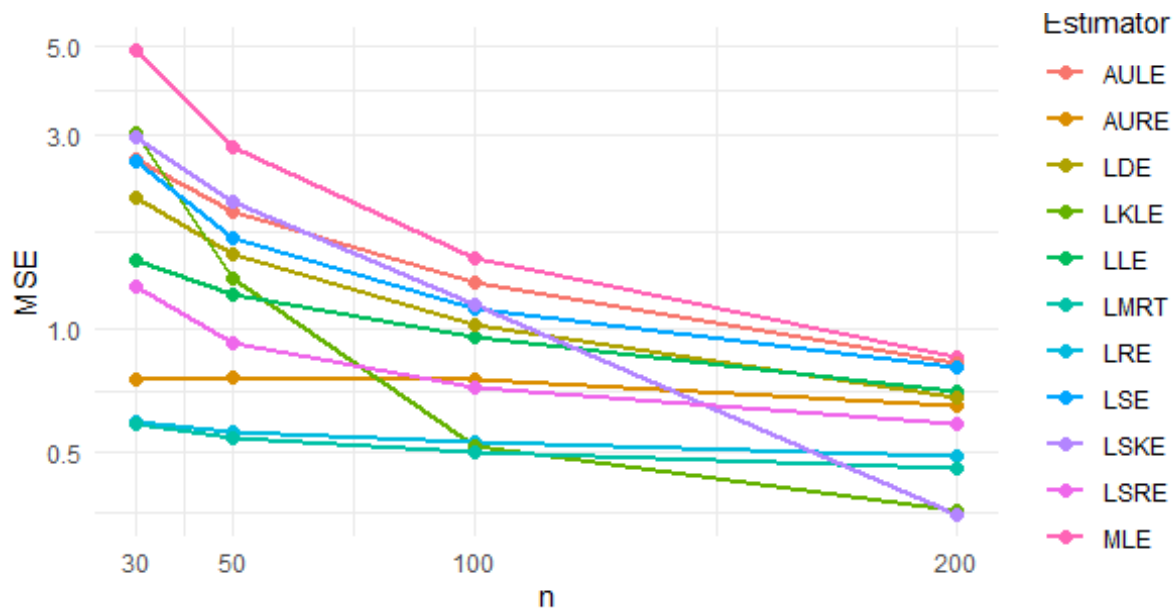


Figure 3.1: Comparing MSE and n for $p = 3$ and $\rho = 0.90$

Interpretation

Figure 3.1 Performance of the estimators as function of n : As the sample size n increases, the MSE decreases for all estimators, indicating improved estimation accuracy with larger samples. The MLE shows relatively high MSE, especially for small n , reflecting its sensitivity to multicollinearity. In contrast, shrinkage estimators such as LRE, LMRT, and LDE consistently yield lower MSE across all sample sizes, demonstrating superior stability and efficiency.

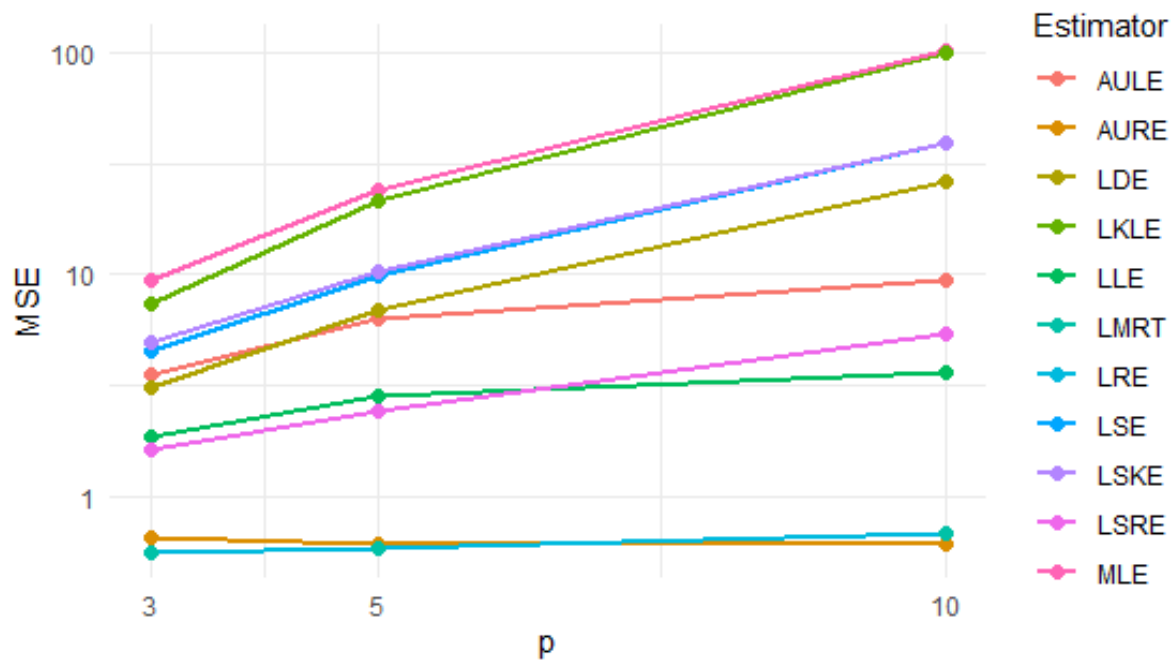


Figure 3.2: Comparing MSE and p for $n = 30$ and $\rho = 0.95$.

Figure 3.2 erformance of the estimators as function of p The figure shows that MSE increases with the number of predictors p for most estimators, particularly MLE, AULE, LSE, and LKLE, highlighting their vulnerability to increasing dimensionality. However, shrinkage-based estimators like LRE, LMRT, and LSRE exhibit relatively stable MSE, indicating robustness against high dimensionality under multicollinearity.

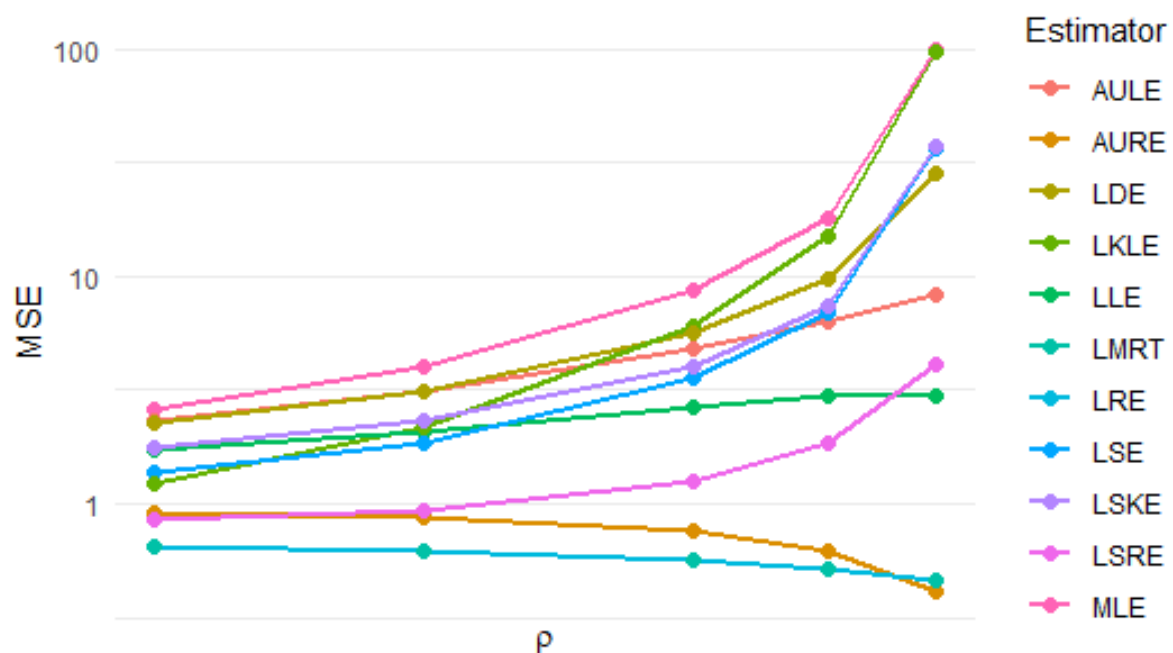


Figure 3.3: Comparing MSE and p for $n = 100$ and $p = 10$.

Interpretation

As the correlation parameter ρ increases, MSE rises for nearly all estimators, reflecting the adverse effect of severe multicollinearity. The increase is especially sharp for MLE, LKLE, and LSKE, indicating strong numerical instability. In contrast, shrinkage estimators such as LRE, LMRT, and LDE maintain comparatively low MSE, demonstrating better resistance to multicollinearity.

Chapter 4

Implementation and Application

4.1 R package Implementation

The proposed biased logistic regression estimators and the associated simulation framework were implemented in the R statistical environment through a custom-developed R package created as part of this study. The package provides modular implementations of several multicollinearity-corrected logistic regression estimators, with each estimator defined in a separate function to ensure clarity and consistency with the theoretical formulations presented in this work.

The package structure follows standard R package conventions and includes dedicated files for individual estimators, generic functions, and supporting utilities. This modular design allows each estimator to be directly linked to its corresponding estimating equations and biasing mechanisms discussed in the theoretical sections of the dissertation.

Cross-validation procedures for selecting the biasing parameters were implemented using user-defined R functions within the package. These functions enable data-driven tuning of the parameters to minimize the Mean Squared Error, thereby ensuring stable and reproducible estimation results. Overall, the package facilitates efficient computation, reproducibility, and systematic comparison of logistic regression estimators under multicollinearity.

Limitations and Future Scope

While the developed R package provides a comprehensive implementation of biased logistic regression estimators, certain extensions can further enhance its usability and reproducibility. In particular, additional documentation and user-level descriptions can be incorporated to make the package more accessible to applied users.

Future work may include the development of detailed simulation and analysis scripts that reproduce all numerical results and figures presented in this study directly from the package. The inclusion of vignettes illustrating theoretical background, parameter tuning via cross-validation, and real-data applications would further strengthen the link between theory and implementation.

Moreover, the package can be extended to handle high-dimensional logistic regression settings and alternative regularization schemes. Such enhancements would improve the applicability of the proposed methods and support further research on multicollinearity in logistic regression models.

The R package developed for this study is available on GitHub at: <https://github.com/Aditya-k-pardeshi/LogisticEstimators>.

4.2 Application

In this section, we consider the prostate cancer dataset from [Kutner \[2005\]](#)

Variable	Description
y	Seminal vesicle invasion (1 = presence, 0 = absence)
x_1	Cancer volume
x_2	Patient weight
x_3	Age of the patient
x_4	Benign prostate hyperplasia
x_5	Capsular penetration
x_6	Gleason score
x_7	Prostate-specific antigen (PSA) level

To assess multicollinearity among the regressors, the eigen values of the matrix $X' \hat{W} X$ are computed using R function. The resulting eigenvalues are

- $\lambda_1 = 69109.31254$
- $\lambda_2 = 58.42260$
- $\lambda_3 = 35.33693$
- $\lambda_4 = 13.64300$
- $\lambda_5 = 10.92500$
- $\lambda_6 = 5.07500$
- $\lambda_7 = 2.50087$

Based on the eigenvalues, the condition index is computed as

$$CI = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}} = \sqrt{\frac{69109.31254}{2.50087}} \approx 166.24.$$

Since this value is much larger than the commonly used threshold of 100, it clearly indicates the presence of severe multicollinearity among the explanatory variables.

Table 3.7 the estimated presents the estimated coefficients and MSE values for different estimators. The coefficients are obtained using Equations of each estimators and the scalar MSE values of these estimators are calculated using their corresponding MSE equations. Parameters have estimated by performing cross validation in R.

Table 4.1: Estimated coefficients and MSEs for the prostate cancer data

Estimator	x_1	x_2	x_3	x_4	x_5	x_6	x_7	MSE
MLE	-0.00041	-0.31977	-0.00263	0.01084	0.21367	-0.23406	0.14794	0.80715
LRE	0.05394	-0.11081	-0.02092	0.01938	0.16602	-0.15281	0.05320	0.20379
LLE	0.02051	-0.24283	-0.00752	0.01257	0.20589	-0.22161	0.11663	0.52857
AURE	0.04374	-0.17504	-0.01174	0.01569	0.20186	-0.21160	0.08658	0.34984
AULE	0.00490	-0.29864	-0.00347	0.01074	0.21400	-0.23541	0.13976	0.73215
LSE	-0.00009	-0.06963	-0.00057	0.00236	0.04653	-0.05097	0.03222	0.17577
LKLE	0.00830	-0.09815	-0.03922	0.02792	0.11837	-0.07157	-0.02155	0.29746
LDE	-0.00333	-0.31944	-0.00264	0.01085	0.21365	-0.23403	0.14781	0.80583
LMRT	0.05395	-0.11078	-0.02093	0.01938	0.16600	-0.15279	0.05319	0.20375
LSRE	0.02694	-0.05534	-0.01045	0.00968	0.08291	-0.07631	0.03156	0.14714
LSKE	0.00490	0.00444	-0.00177	0.00126	0.00535	-0.00324	0.00097	0.21927

Interpretation

Table 3.7 displays the estimated regression coefficients and MSE values for different estimators applied to the prostate cancer dataset. It is evident that the shrinkage-based estimators outperform the classical MLE in terms of Mean Squared Error (MSE). Among all estimators, LSRE yields the smallest MSE value (0.14714), indicating the most accurate and stable estimation for the prostate cancer data. This is followed closely by LSE, LMRT, and LRE, all of which show substantially lower MSE values compared to MLE.

The classical MLE and estimators such as LDE and AULE exhibit relatively large MSE values, reflecting their sensitivity to multicollinearity present in the data. In contrast, estimators like LSRE, LMRT, and LRE effectively reduce estimation error by incorporating shrinkage, leading to improved numerical stability.

Overall, the results clearly indicate that LSRE, followed by LMRT and LRE, are the best-performing estimators for the prostate cancer dataset, making them more reliable alternatives to the classical MLE in the presence of multicollinearity.

4.3 Overall Conclusion

Multicollinearity is a serious problem in logistic regression as it leads to unstable parameter estimates and inflated Mean Squared Error (MSE) for the classical Maximum Likelihood Estimator (MLE). In this study, several biased and shrinkage logistic regression estimators were examined theoretically, through simulation, and using a real prostate cancer dataset.

The simulation results clearly demonstrate that the performance of MLE deteriorates as the correlation among predictors increases and as the number of explanatory variables grows. In contrast, shrinkage-based estimators consistently produce lower MSE values across different sample sizes, correlation levels, and model dimensions. Among all estimators considered, LRE, LMRT, and LSRE show superior and stable performance under severe multicollinearity.

The real data application using the prostate cancer dataset further supports the simulation findings. The large condition index confirms the presence of severe multicollinearity in the data. In this setting, shrinkage estimators, particularly LSRE, LMRT, and LRE, yield substantially smaller MSE values compared to MLE, indicating improved estimation accuracy and numerical stability.

Overall, the results suggest that shrinkage-based logistic regression estimators are more reliable alternatives to the classical MLE when multicollinearity is present.

Appendix

R Code for Simulation Study

This appendix contains the complete R code used to perform the simulation study, compute the Mean Squared Error (MSE), and generate the figures for comparing logistic regression estimators under multicollinearity.

Listing 4.1: R code for simulation and MSE comparison of logistic regression estimators

```
1 library(MASS)
2 library(readxl)
3 library(ggplot2)
4 library(dplyr)
5 library(tidyr)
6 library(patchwork)
7
8 # Generate explanatory variables
9 generate_explanatory_vars <- function(n, p, rho) {
10   Sigma <- matrix(rho, nrow = p, ncol = p)
11   diag(Sigma) <- 1
12   MASS::mvrnorm(n, mu = rep(0, p), Sigma = Sigma)
13 }
14
15 # Generate response
16 generate_response <- function(X, beta) {
17   eta <- X %*% beta
18   p <- 1 / (1 + exp(-eta))
19   rbinom(nrow(X), 1, p)
20 }
21
22 # Helper: get MLE fit
23 get_mle <- function(X, y) {
24   model <- glm(y ~ ., data = data.frame(y, X), family = binomial)
25   beta_mle <- coef(model)
26   X_design <- model.matrix(model)
27   p_hat <- fitted(model)
28   W <- diag(p_hat * (1 - p_hat))
29   W_inv <- ginv(W)
30   z <- X_design %*% beta_mle + W_inv %*% (y - p_hat)
31   return(list(beta_mle = beta_mle, X = X_design, W = W, z = z))
32 }
33
34 # Logistic Ridge Estimator
35 ridge_estimator <- function(X, y, k) {
36   mle <- get_mle(X, y)
37   XWX <- t(mle$X) %*% mle$W %*% mle$X
38   A <- solve(XWX + k * diag(ncol(XWX))) %*% XWX
```

```

39   drop(A %*% mle$beta_mle)
40 }
41
42 # Logistic Liu Estimator
43 liu_estimator <- function(X, y, d) {
44   mle <- get_mle(X, y)
45   XWX <- t(mle$X) %*% mle$W %*% mle$X
46   A <- solve(XWX + diag(ncol(XWX))) %*% (XWX + d * diag(ncol(XWX)))
47   drop(A %*% mle$beta_mle)
48 }
49
50
51 # 3. AURE - Almost Unbiased Ridge Estimator
52 aure_estimator <- function(X, y, k) {
53   mle <- get_mle(X, y)
54   XWX <- t(mle$X) %*% mle$W %*% mle$X
55   A <- diag(ncol(XWX)) - k^2 * ginv((XWX + k * diag(ncol(XWX)))^2)
56   beta <- A %*% mle$beta_mle
57   return(drop(beta))
58 }
59
60 # 4. AULE - Almost Unbiased Liu Estimator
61 aulue_estimator <- function(X, y, d) {
62   mle <- get_mle(X, y)
63   XWX <- t(mle$X) %*% mle$W %*% mle$X
64   A <- diag(ncol(XWX)) - (1 - d)^2 * solve((XWX + diag(ncol(XWX)))^2)
65   beta <- A %*% mle$beta_mle
66   return(drop(beta))
67 }
68
69 # 5. ALLE - Adjusted Liu Estimator
70 alle_estimator <- function(X, y, d) {
71   mle <- get_mle(X, y)
72   XWX <- t(mle$X) %*% mle$W %*% mle$X
73   A <- solve(XWX + diag(ncol(XWX))) %*% (XWX - d * diag(ncol(XWX)))
74   beta <- A %*% mle$beta_mle
75   return(drop(beta))
76 }
77
78 # 6. LKLE - Kibria Logistic Estimator
79 lkle_estimator <- function(X, y, k) {
80   mle <- get_mle(X, y)
81   XWX <- t(mle$X) %*% mle$W %*% mle$X
82   K <- solve(XWX + k * diag(ncol(XWX))) %*% (XWX - k * diag(ncol(XWX)))
83   beta <- K %*% mle$beta_mle
84   return(drop(beta))
85 }
86
87 # 7. LDE - Dorugade Estimator
88 lde_estimator <- function(X, y, k, d) {
89   mle <- get_mle(X, y)
90   XWX <- t(mle$X) %*% mle$W %*% mle$X
91   A <- solve(XWX + k * d * diag(ncol(XWX))) %*% XWX
92   beta <- A %*% mle$beta_mle
93   return(drop(beta))
94 }
95
96 # 8. LMRT - Modified Ridge Estimator

```



```

97 lmrt_estimator <- function(X, y, k, d) {
98   mle <- get_mle(X, y)
99   XWX <- t(mle$X) %*% mle$W %*% mle$X
100   A <- solve(XWX + k * (1 + d) * diag(ncol(XWX))) %*% XWX
101   beta <- A %*% mle$beta_mle
102   return(drop(beta))
103 }
104
105 # 9. LSRE - Logistic Stein Ridge Estimator
106 lsre_estimator <- function(X, y, k) {
107   mle <- get_mle(X, y)
108   XWX <- t(mle$X) %*% mle$W %*% mle$X
109   A <- solve(XWX + k * diag(ncol(XWX))) %*% XWX
110   num <- t(mle$beta_mle) %*% A %*% mle$beta_mle
111   denom <- t(mle$beta_mle) %*% t(A) %*% A %*% mle$beta_mle + sum(diag(A
112     %*% solve(XWX) %*% t(A)))
113   c_r <- as.numeric(num / denom)
114   beta <- c_r * A %*% mle$beta_mle
115   return(drop(beta))
116 }
117
118 # 10. LSKE - Logistic Stein Kibria Estimator
119 lske_estimator <- function(X, y, k) {
120   mle <- get_mle(X, y)
121   XWX <- t(mle$X) %*% mle$W %*% mle$X
122   K <- solve(XWX + k * diag(ncol(XWX))) %*% (XWX - k * diag(ncol(XWX)))
123   num <- t(mle$beta_mle) %*% K %*% mle$beta_mle
124   denom <- t(mle$beta_mle) %*% t(K) %*% K %*% mle$beta_mle + sum(diag(K
125     %*% solve(XWX) %*% t(K)))
126   c_k <- as.numeric(num / denom)
127   beta <- c_k * K %*% mle$beta_mle
128   return(drop(beta))
129 }
130
131 ests <- cbind(
132   MLE = get_mle(X,y)$beta_mle,
133   Ridge = ridge_estimator(X, y, k=1.6),
134   Liu = liu_estimator(X, y, d=0),
135   AURE = aure_estimator(X, y, k=6),
136   AULE = aule_estimator(X, y, d=0),
137   ALLE = alle_estimator(X, y, d=0),
138   LKLE = lkke_estimator(X, y, k=0.0001),
139   LDE = lde_estimator(X, y, k=0.6, d=2.4),
140   LMRT = lmrt_estimator(X, y, k=0.6, d=1.3),
141   LSRE = lsre_estimator(X, y, k=6),
142   LSKE = lske_estimator(X, y, k=0.001)
143 )
144
145 ests
146
147 simulate_mse_p <- function(p_values, n, rho, R = 500, k, d) {
148   estimator_names <- c("MLE", "Ridge", "Liu", "AURE", "AULE", "ALLE",
149     "LKLE", "LDE", "LMRT", "LSRE", "LSKE")
150   results <- data.frame()
151
152   for (p in p_values) {
153     beta <- rep(1, p) # true coefficient vector
154     mse_mat <- matrix(0, nrow = R, ncol = length(estimator_names))

```

```

153 for (r in 1:R) {
154   # Generate correlated design matrix
155   Sigma <- matrix(rho, nrow = p, ncol = p)
156   diag(Sigma) <- 1
157   X <- MASS::mvrnorm(n, mu = rep(0, p), Sigma = Sigma)
158   eta <- X %*% beta
159   p_i <- 1 / (1 + exp(-eta))
160   y <- rbinom(n, 1, p_i)
161
162   # Compute estimates
163   ests <- cbind(
164     MLE = get_mle(X,y)$beta_mle,
165     Ridge = ridge_estimator(X, y, k=1.6),
166     Liu = liu_estimator(X, y, d=0),
167     AURE = aure_estimator(X, y, k=6),
168     AULE = aulue_estimator(X, y, d=0),
169     ALLE = alle_estimator(X, y, d=0),
170     LKLE = lkle_estimator(X, y, k=0.0001),
171     LDE = lde_estimator(X, y, k=0.6, d=2.4),
172     LMRT = lmrt_estimator(X, y, k=0.6, d=1.3),
173     LSRE = lsre_estimator(X, y, k=6),
174     LSKE = lske_estimator(X, y, k=0.001)
175   )
176
177   for (j in 1:length(estimator_names)) {
178     mse_mat[r, j] <- mean((ests[, j] - beta)^2, na.rm = TRUE)
179   }
180 }
181
182 avg_mse <- setNames(colMeans(mse_mat, na.rm=TRUE), estimator_names)
183
184 results <- rbind(results,
185   data.frame(p = rep(p, length(estimator_names)),
186     Estimator = estimator_names,
187     MSE = as.numeric(avg_mse)))
188 }
189
190 rownames(results) <- NULL
191 return(results)
192 }
193
194 # ---- Run simulation ----
195 set.seed(123)
196 p_values <- c(3, 5, 10)
197 n <- 3
198 rho <- 0.9
199
200 res_p <- simulate_mse_p(p_values, n, rho, R = 2000)
201
202
203
204 # =====
205 # ESTIMATORS (COMMON)
206 # =====
207 estimators <- c("MLE", "LRE", "LLE", "AURE", "AULE",
208   "LSE", "LKLE", "LDE", "LMRT", "LSRE", "LSKE")
209
210 # =====

```

```

211 # PLOT 1: MSE vs n (p = 3, rho = 0.90)
212 # =====
213 mse_n <- data.frame(
214   Estimator = estimators,
215   n30 = c
216     (4.847,0.597,1.486,0.757,2.612,2.584,3.043,2.108,0.587,1.277,2.966)
217   ,
218   n50 = c
219     (2.795,0.559,1.223,0.767,1.950,1.678,1.329,1.533,0.541,0.931,2.058)
220   ,
221   n100 = c
222     (1.503,0.528,0.961,0.753,1.310,1.129,0.520,1.022,0.500,0.720,1.157)
223   ,
224   n200 = c
225     (0.859,0.487,0.706,0.647,0.828,0.813,0.358,0.684,0.459,0.590,0.353)
226 )
227
228 mse_n_long <- mse_n %>%
229   pivot_longer(-Estimator, names_to = "n", values_to = "MSE") %>%
230   mutate(n = as.numeric(sub("n", "", n)))
231
232 p1 <- ggplot(mse_n_long, aes(n, MSE, color = Estimator)) +
233   geom_line(linewidth = 0.9) +
234   geom_point(size = 2) +
235   scale_x_continuous(breaks = c(30,50,100,200)) +
236   labs(x = "n", y = "MSE") +
237   scale_y_log10() +
238   theme_minimal()
239
240 # =====
241 # PLOT 2: MSE vs p (n = 30, rho = 0.95)
242 # =====
243 mse_p <- data.frame(
244   Estimator = estimators,
245   p3 = c
246     (9.517,0.562,1.844,0.654,3.585,4.575,7.315,3.092,0.559,1.646,4.954)
247   ,
248   p5 = c
249     (24.017,0.589,2.856,0.617,6.308,9.924,21.275,6.867,0.591,2.460,10.297)
250   ,
251   p10 = c
252     (102.534,0.679,3.645,0.619,9.489,38.973,99.471,26.366,0.681,5.392,39.261)
253 )
254
255 mse_p_long <- mse_p %>%
256   pivot_longer(-Estimator, names_to = "p", values_to = "MSE") %>%
257   mutate(p = as.numeric(sub("p", "", p)))
258
259 p2 <- ggplot(mse_p_long, aes(p, MSE, color = Estimator)) +
260   geom_line(linewidth = 0.9) +
261   geom_point(size = 2) +
262   scale_x_continuous(breaks = c(3,5,10)) +
263   labs(x = "p", y = "MSE") +
264   scale_y_log10() +

```

```

255   theme_minimal()
256
257 # =====
258 # PLOT 3: MSE vs rho (n = 100, p = 10)
259 # =====
260 mse_rho <- data.frame(
261   Estimator = estimators,
262   rho_07    = c
263     (2.634,0.657,1.730,0.916,2.347,1.391,1.217,2.263,0.652,0.849,1.788)
264   ,
265   rho_08    = c
266     (4.003,0.625,2.075,0.878,3.140,1.858,2.162,3.143,0.620,0.939,2.299)
267   ,
268   rho_09    = c
269     (8.666,0.566,2.641,0.759,4.792,3.556,6.098,5.700,0.563,1.265,4.039)
270   ,
271   rho_095   = c
272     (18.173,0.522,2.957,0.621,6.343,6.939,15.117,9.804,0.521,1.852,7.429)
273   ,
274   rho_099   = c
275     (99.612,0.459,2.986,0.408,8.392,36.432,96.042,28.239,0.462,4.076,36.955)
276 )
277
278 mse_rho_long <- mse_rho %>%
279   pivot_longer(-Estimator, names_to = "rho", values_to = "MSE") %>%
280   mutate(rho = as.numeric(sub("rho_", "0.", rho)))
281
282 p3 <- ggplot(mse_rho_long, aes(rho, MSE, color = Estimator)) +
283   geom_line(linewidth = 0.9) +
284   geom_point(size = 2) +
285   scale_x_continuous(breaks = c(0.7,0.8,0.9,0.95,0.99)) +
286   labs(x = expression(rho), y = "MSE") +
287   scale_y_log10() +
288   theme_minimal()
289
290 # (Remaining estimator functions and plotting code continue here)

```

Bibliography

- M. Amin, M. N. Akram, B. G. Kibria, H. M. Alshanbari, N. Fatima, and A. Elhassanein. On the estimation of the binary response model. *Axioms*, 12(2): 175, 2023.
- X. Chang. On the almost unbiased ridge and liu estimator in the logistic regression model. In *2015 International Conference on Social Science, Education Management and Sports Education*, pages 1658–1660. Atlantis Press, 2015.
- D. G. Gibbons. A simulation study of some ridge estimators. *Journal of the American Statistical Association*, 76(373):131–139, 1981.
- A. E. Hoerl and R. W. Kennard. Ridge regression—1980: Advances, algorithms, and applications. *American Journal of Mathematical and Management Sciences*, 1(1):5–83, 1981.
- M. A. Hoque and B. G. Kibria. Performance of some estimators for the multicollinear logistic regression model: theory, simulation, and applications. *Research in Statistics*, 2(1):2364747, 2024.
- K. Kadiyala. A class of almost unbiased and efficient estimators of regression coefficients. *Economics Letters*, 16(3-4):293–296, 1984.
- L. Kejian. A new class of biased estimate in linear regression. *Communications in Statistics-Theory and Methods*, 22(2):393–402, 1993.
- M. H. Kutner. Applied linear statistical models. 2005.
- A. F. Lukman, B. G. Kibria, C. K. Nziku, M. Amin, E. T. Adewuyi, and R. Farghali. KI estimator: Dealing with multicollinearity in the logistic regression model. *Mathematics*, 11(2):340, 2023.
- K. Månsson, B. G. Kibria, and G. Shukur. On liu estimators for the logit regression model. *Economic Modelling*, 29(4):1483–1488, 2012.

- M. Matin, A. Saleh, and E. Md. Some improved estimators in logistic regression model. *Journal of Statistical Research*, 39(0):2, 2005.
- R. H. Mayers. Classical and modern regression with applications: PwsKent publ. Co.: Boston, 1990.
- G. C. McDonald and D. I. Galarneau. A monte carlo evaluation of some ridge-type estimators. *Journal of the American Statistical Association*, 70(350): 407–416, 1975.
- S. Menard. *Applied logistic regression analysis*. SAGE publications, 2001.
- H. Midi, S. K. Sarkar, and S. Rana. Collinearity diagnostics of binary logistic regression model. *Journal of interdisciplinary mathematics*, 13(3):253–267, 2010.
- S. Rana, H. Midi, and S. Sarkar. Validation and performance analysis of binary logistic regression model. In *Proceedings of the WSEAS International Conference on Environmental, Medicine and Health Sciences*, pages 23–25. WSEAS Press, 2010.
- R. Schaefer, L. Roi, and R. Wolfe. A ridge logistic estimator. *Communications in Statistics-Theory and Methods*, 13(1):99–113, 1984.
- N. N. Urgan and M. Tez. Liu estimator in logistic regression when the data are collinear. In *20th euro mini conference*, pages 323–327, 2008.